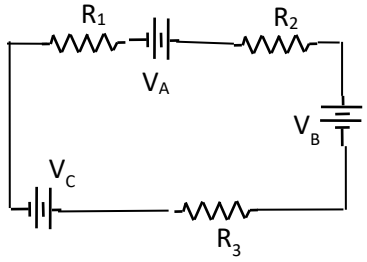
Physics 2 Problem Solving – Chapter 43 – Kirchhoff's rules problems

1) Use Kirchhoff's rules to solve for the current in the circuit.

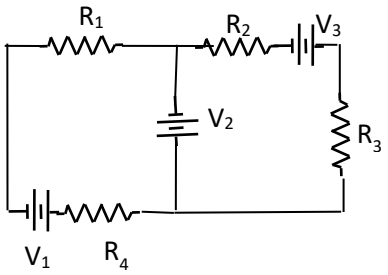
Answer: 1 A



V_A = 20 V
V_B = 30 V
V_C = 20 Ω
R_1 = 10 Ω
R_2 = 15 Ω
R_3 = 5 Ω

2) Use Kirchhoff's rules to solve for the current through each of the resistors R_1, R_2 and V_2.

Answer: I_1 = 0.33 A, I_3 = 0.6 A and I_2 = 0.27 A



V_1 = 100 V
V_2 = 200 V
V_3 = 50 V
R_1 = 100 Ω
R_2 = 200 Ω
R_3 = 50 Ω
R_4 = 200 Ω